Restaurant Tips Prediction Using Excel

Project Overview

This project uses Microsoft Excel to build a predictive model that estimates restaurant tips based on customer and dining information. The dataset was cleaned, transformed, and analyzed using Excel functions and the Data Analysis ToolPak to identify the factors that influence tipping behavior.

The project demonstrates the complete data analytics workflow, from data cleaning and preprocessing to regression modeling and model evaluation, without using programming languages.

Objectives

- Clean and prepare the restaurant tips dataset.
- Identify independent and dependent variables.
- Convert categorical variables into numeric values using Excel IF functions.
- Analyze relationships between variables using the CORREL function.
- Build a multiple linear regression model to predict customer tips.
- Compare actual and predicted tip values.
- Evaluate model performance using Root Mean Square Error (RMSE).

Tools Used

- Microsoft Excel
- Excel Functions (IF, CORREL, AVERAGE, SQRT)
- Data Analysis ToolPak (Regression)
- PivotTables
- Charts and Data Visualization

Project Workflow

- Imported and cleaned the dataset by removing missing values and duplicate records.
- Identified **Tip** as the dependent variable and customer/dining information as independent variables.
- Encoded categorical variables (Sex, Smoker, Day, and Time) into numeric values.

- Calculated correlation coefficients to determine which variables were most related to tip amount.
- Built a multiple linear regression model using Excel's Data Analysis ToolPak.
- Generated predicted tip values using the regression equation.
- Compared predicted values with actual tips.
- Calculated RMSE to measure prediction accuracy.
- Created visualizations including:
    - Actual vs Predicted Tips
    - Total Bill vs Tip Scatter Plot
    - Correlation Analysis
    - Average Tips by Day
    - Average Tips by Time
    - Average Tips by Gender

Key Skills Demonstrated

- Data Cleaning
- Data Transformation
- Feature Engineering
- Correlation Analysis
- Regression Analysis
- Predictive Analytics
- Model Evaluation
- Dashboard Design
- Data Visualization
- Excel Analytics

Results

The regression model successfully predicted restaurant tip amounts using customer and dining characteristics. Correlation analysis showed which variables had the greatest influence on tipping behavior, while the RMSE metric was used to evaluate the model's prediction accuracy. The project highlights how Microsoft Excel can be used to perform end-to-end predictive analytics without requiring programming.